

Jahanzaib Khan

Personal Information

Address Dr.-Gessler Strasse. 5, 93051 Regensburg, Germany
Contact +4917637092163
Email jahanzaibkhan208@gmail.com
Date of Birth 30.01.1995
Nationality Pakistan

Core Skills

Programming C#, C/C++, Python, Embedded C, MATLAB

Software & Tools .NET, WPF (MVVM), Visual Studio, Git, Jira, Confluence, PyCharm, Eclipse, MSSQL, AUTOCAD, KiCAD, Microwind, MS Office

Embedded & Hardware Tiva C (TM4C123), Arduino, SPI, I2C, UART, PWM, IMU Integration, Stepper Motor Control

FPGA & ASIC Verilog HDL, OpenFPGA, VPR, Yosys, OpenROAD, LibreLane, RTL- GDSII, PDN, Static Timing Analysis(STA), Floorplaning and Placement

Development Concepts OOP, Multithreading, Asynchronous Programming, Debugging, Logging

Automation Python Scripting, Unit Testing, XML Processing

PCB & Design PCB Design, Circuit Design, Sensor Integration

Automation Basic knowledge of PLC (Delta PLC), Ladder Logic (LD), industrial automation concepts

Communication TCP/IP, VISA, SCPI, Serial Communication

Instruments Multimeter, Oscilloscope, Function Generator, Vector Network Analyzer(VNA), Logic Analyzer

Languages English (C1), German (A2), Urdu (Native), Hindi (Fluent)

Work Experience

- Oct 2025–March 2026 **Custom FPGA Design using OpenFPGA and Librelane, OTH Regensburg, Regensburg, Germany**
- Designed a custom 4×4 FPGA architecture (LUT4 CLBs, routing, interconnect) using OpenFPGA
 - Developed Verilog RTL for configurable logic and routing blocks
 - Performed logic synthesis and RTL-to-GDSII flow using Yosys and LibreLane
 - Executed physical design (floorplanning, placement, routing) using OpenROAD
 - Conducted functional verification and timing analysis, resolving PPA and connectivity issues
- Oct 2023–Feb 2024 **Software Developer in Tool Development, Vitesco Technologies, Regensburg, Germany**
- Developed Python automation and unit-test scripts, increasing test coverage by 35%
 - Built automation tools (Python/Batch) for workflows such as HTML-to-PDF conversion and XML processing
 - Designed desktop applications using C#, WPF (.NET) with MVVM architecture
 - Implemented network communication and system automation (TCP/IP, server checks, file handling)

- April **Application Developer, Keysight Technologies, Boblingen, Germany**
- 2023–Sep 2023
- Refactored a legacy C# WPF application to MVVM, improving maintainability and testability
 - Enhanced GUI and optimized data binding for improved usability and performance
 - Implemented input validation, exception handling, and logging for reliable processing
 - Developed multithreaded and asynchronous solutions for efficient execution of measurement tasks
 - Automated measurement workflows and integrated SCPI-based instrument communication (TCP/IP, VISA)

Projects

- Jan 2026–present **Thermal Sensor Data Analysis System, OTH Regensburg, Regensburg, Germany**
- Developed an embedded system for thermal sensor analysis with full hardware–software integration
 - Implemented embedded C on Tiva C for stepper motor control and IMU interfacing
 - Designed MATLAB GUI for motor control and real-time communication
 - Designed and assembled a custom PCB integrating IMU and thermal sensor
 - Performed system testing and validation of the complete solution
- Sep 2017–Sep 2018 **Final Year Project – Cricket Bowling Machine, Usman Institute of Technology, Karachi, Pakistan**
- Designed and developed the mechanical structure of a cricket bowling machine for controlled ball delivery
 - Implemented C++-based Arduino firmware to control a BLDC motor using PWM for variable speed regulation
 - Developed a keypad-based user interface to adjust motor speed and control operation modes
 - Integrated an LCD display to provide real-time feedback on system status and speed
- sep 2017 **PLC-Based Automation (University Lab), Usman Institute Of Technology, Karachi, Pakistan**
- Developed basic control programs using Ladder Logic on **PLC** (Delta PLC)
 - Implemented simple automation tasks integrating sensors and actuators
 - Performed testing and debugging of control logic in a lab environment
 - Gained practical understanding of industrial control systems and PLC I/O interfacing

Courses

- April 2024 Fundamentals of Linux Linkedin Learning
- March 2024 Python Bootcamp Udemy

Education

- 2021–2026 **Master of Science in Electrical and Microsystem Engineering, OTH Regensburg, Regensburg, Germany, Grade: 2.2**
- Courses:** Multicore and Multiprocessing, Advanced Engineering Mathematics, Electronics Product Engineering
- Projects:**
- Design Fire Alarm system in c and using keil for debugging
 - Implementation Trapezoidal Rule in C++ and Matlab
- 2014–2018 **Bachelor of Engineering in Electronics Engineering, Usman Institute of Technology, Karachi, Pakistan, Grade: 1.6**
- Courses:** Electronic Circuit Design, Electronics Network Analysis, Microprocessor, Digital Logic Design, Power Electronics, VLSI
- Projects:**
- Designed and tested ADC and DAC circuits using ICs
 - Built a portable speaker with AUX input, volume control, and amplifier IC
 - Implemented Output control using encoder/decoder logic with keypad input
 - Designed and built a 0–12V variable power supply with adjustable voltage output